General-Purpose Technologies and Stock Market Bubbles[1]

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Objectives

1. Formal connection between general-purpose technologies and stock price bubbles
2. Bubble necessity despite rationally anticipated collapse
3. Real effects of bubble-backed IPO proceeds and collapse

Technological innovation and bubbles

- Many asset-price booms coincide with major technological innovations (Scheinkman, 2014; Quinn and Turner, 2020)
- Examples:
  - 1720s Mississippi and South Sea bubbles: Atlantic trade, insurance
  - 1840s Railway Mania: steam engine, railway network
  - 1890s Bicycle Mania: pneumatic tire
  - 1920s US stock boom: electricity, automobiles, radio
  - 1990s dot-com bubble: Internet; today: AI?
- General-purpose technologies (GPTs) (Bresnahan and Trajtenberg, 1995):
  - (pervasiveness) useful across many sectors
  - (improvement) improve over time
  - (innovation spanning) make it easier to develop further innovations

Rational bubble despite anticipated collapse

- Bubble: asset price (Q) $>$ fundamental value (V)
  - $V =$ expected present discounted value of dividends (D)
- "Rational bubble": $Q > V$ despite rational agents with common beliefs and information (Hirano and Toda, 2024)
- Standard difficulties:
  - Dividends tighten no-bubble condition relative to pure bubble ($D \equiv 0$)
  - Known collapse date rules out bubbles by backward induction
- Can bubble be necessary despite anticipated complete collapse?

This paper

- Macro-finance model of innovation and stock price bubble
- Skilled agents choose between production and R&D
- R&D creates new firms whose shares pay dividends
- Knowledge spillovers differ across production factors
    - Emerging GPT: unbalanced growth
    - Mature GPT: absorbing balanced-growth state
- Agents rationally anticipate regime transition and bubble collapse

Main results

1. Bubble necessity despite anticipated collapse
   - Unique equilibrium has $Q_t > V_t$ during unbalanced growth
   - Survival probability affects duration, not existence
2. Bubble-backed innovation and regime transition
   - Capitalizing bubble into IPO proceeds increases R&D employment
   - Later collapse implies larger contraction but more inherited knowledge
3. Balanced growth: knife-edge
   - Restrictions generating balanced growth also rule out bubble by construction

Bezos

Related literature

- Rational bubbles: Samuelson (1958), Bewley (1980), Tirole (1985), Scheinkman and Weiss (1986), Kocherlakota (1992), and Santos and Woodford (1997)
- Bubbles attached to dividend-paying assets: Hirano and Toda (2024, 2025a,b) and Sorger (2026)
- Collapsing bubbles: Blanchard (1979), Weil (1987), Michel and Wigniolle (2003), and Guerron-Quintana et al. (2023)
- Technological innovation and asset booms: Olivier (2000), Zeira (1999), Pástor and Veronesi (2009), and Haddad et al. (2022)

Endowment economy

- Two-period OLG, $t = 0, 1, \dots$
- Unit mass of young agents born each date
- Young receive endowment $e_t > 0$, old receive none
- Initial old own one unit of long-lived asset
  - ex-dividend price $Q_t > 0$
  - dividend $D_t > 0$
- Complete markets with Arrow securities in zero net supply
- Adapted stochastic process (e_t, D_t)
- Utility is log Epstein–Zin ($\mathrm{EIS} = 1$, $\beta \in (0, 1)$, $\gamma \geq 0$):

$$U(c_t^y, c_{t+1}^o) = (1 - \beta) \log c_t^y + \beta \log \mathrm{E}_t[(c_{t+1}^o)^{1-\gamma}]^{\frac{1}{1-\gamma}}$$

Equilibrium

- Generation-t budget constraints:

$$\begin{aligned} c_t^y + Q_t\theta_t &= e_t, \\ c_{t+1}^o &= (Q_{t+1} + D_{t+1})\theta_t \end{aligned}$$

- Market clearing:

$$\begin{aligned} c_t^y + c_t^o &= e_t + D_t, \\ \theta_t &= 1 \end{aligned}$$

- Unit EIS implies $c_t^y = (1-\beta)e_t$

Solution

Unique equilibrium

- Young's budget constraint and asset-market clearing imply

$$Q_t = e_t - c_t^y = \beta e_t$$

Proposition

Unique rational expectations equilibrium exists, with asset price

$$Q_t = \beta e_t,$$

and consumption

$$(c_t^y, c_t^o) = ((1-\beta)e_t, \beta e_t + D_t).$$

Fundamental value and bubble

- Let one-period SDF be $m_{t\to t+1}$,
  $m_{t\to T} := m_{t\to t+1} \times \cdots \times m_{T-1\to T}$ ▸ SDF
- Euler equation:

$$Q_t = \mathrm{E}_t[m_{t\to t+1}(Q_{t+1} + D_{t+1})]$$

- Iteration and monotone convergence yield

$$Q_t = \underbrace{\mathrm{E}_t\left[\sum_{s=t+1}^{\infty} m_{t\to s} D_s\right]}_{=:V_t} + \underbrace{\lim_{T\to\infty} \mathrm{E}_t[m_{t\to T} Q_T]}_{=:B_t}$$

- Hence $Q_t = V_t$ if and only if no-bubble condition holds:

$$\lim_{T\to\infty} \mathrm{E}_t[m_{t\to T} Q_T] = 0$$

Temporary unbalanced growth

Assumption

Two states $z_t \in \{u, b\}$ with transition probabilities

$$\Pr(z_{t+1} = u \mid z_t = u) = \pi \in (0, 1),$$
$$\Pr(z_{t+1} = b \mid z_t = b) = 1.$$

- u: endowments and dividends may grow at different rates
- b: absorbing balanced-growth state
- First transition date $\tau = \min \{t : z_t = b\}$ geometrically distributed

State b: balanced growth

Assumption

If state b reached at τ, all subsequent endowments and dividends are deterministic, with

$$\frac{e_{t+1}}{e_t} = \frac{D_{t+1}}{D_t}, \qquad t \geq \tau.$$

Proposition

After transition to state b, $Q_t = V_t$ for all $t \geq \tau$.

- With common gross growth factor G, Q_t grows at G, whereas

$$R = \frac{Q_{t+1} + D_{t+1}}{Q_t} = G\frac{\beta e_0 + D_0}{\beta e_0} > G$$

- Discounted terminal price therefore vanishes

Conditional aggregate uncertainty

Assumption

For every $t \geq 1$, past regime history $(z_0, \ldots, z_{t-1})$ determines two possible pairs,

$$(e_t^u, D_t^u) \qquad \textit{and} \qquad (e_t^b, D_t^b).$$

Current state selects realized pair:

$$(e_t, D_t) = (e_t^{z_t}, D_t^{z_t}).$$

No other aggregate uncertainty.

Thus z_t is only new aggregate information at date t

Bubble-sustaining paths

- Decompose discounted terminal price by transition date:

$$\begin{aligned} \mathrm{E}_0[m_{0\to t} Q_t] = & \sum_{j=1}^{t} \pi^{j-1}(1-\pi)\,\mathrm{E}_0[m_{0\to t} Q_t \mid \tau = j] \\ & + \pi^t\,\mathrm{E}_0[m_{0\to t} Q_t \mid \tau > t] \end{aligned}$$

- State b is bubbleless, so bubble exists if and only if

$$\lim_{t\to\infty} \pi^t\,\mathrm{E}_0[m_{0\to t} Q_t \mid \tau > t] > 0$$

- Only state-u histories sustain bubble

Bubble characterization under anticipated collapse

Let $\Pi_t^z := Q_t^z + D_t^z$ denote stock payoff conditional on state $z \in \{u, b\}$

Theorem

Under Epstein–Zin preferences with unit EIS, regime-transition assumption, absorbing balanced-growth assumption, and conditional aggregate uncertainty, if $z_0 = u$, bubble exists at $t = 0$ if and only if

$$\text{Vanishing dividend yield:} \quad \sum_{t=1}^{\infty} \frac{D_t^u}{Q_t^u} < \infty,$$

$$\text{Large transition-state decline:} \quad \sum_{t=1}^{\infty} \left(\frac{\Pi_t^b}{\Pi_t^u} \right)^{1-\gamma} < \infty.$$

▸ Proof

Economic interpretation

- Suppose $\gamma < 1$
  1. During state u: dividend yield tends to zero, price-dividend ratio to infinity
  2. At transition, consumption and stock payoff fall relative to continuation path
  3. Payoff decline dominates increased marginal valuation
- Bubble survives state u only if eventual collapse becomes sufficiently severe
  - $\gamma = 1$: each second-series term equals one
  - $\gamma > 1$: transition-state terms diverge when $\Pi_t^b / \Pi_t^u \to 0$
- Similar result with general Epstein–Zin ▸ Details

Bubble necessity

- Unique equilibrium asset price
- Under both summability conditions, expected-to-collapse bubble is unique equilibrium outcome
- Conditions independent of survival probability π
  - π governs expected duration, not existence
- Unlike our result, stochastic pure bubble in Weil (1987) requires sufficiently high survival probability and yields equilibrium indeterminacy
- Complete collapse requires aggregate news at transition
  - If (e_t, D_t) known at $t-1$, no bubble

Model with innovation and intangible capital

- Variety-expansion model with monopolistic competition (Grossman and Helpman, 1991)
- Mass $L > 0$ of unskilled agents
  - work in final-good sector
- Mass $H > 0$ of skilled agents
  - produce knowledge-intensive intermediate goods, or
  - conduct R&D and establish new firms
- Each intermediate-good firm issues one share and pays monopoly profits as dividends
- n_t: variety count, knowledge stock, and stock supply

Timing and ownership

- At date t:
  1. Productivities (A_{Xt}, A_{Lt}) are realized
  2. Labor supply and skilled-agent choice between production and R&D
  3. Production and R&D take place
  4. Incumbent firms pay dividends to old shareholders
  5. Innovators create $n_{t+1} - n_t$ new firms and issue shares through IPOs
  6. Old shareholders sell incumbent shares, young agents buy all n_{t+1} shares
- Incumbent and newly issued shares are identical ex-dividend claims satisfying law of one price

Consumption good sector

- Competitive final-good firm produces

$$
\begin{aligned}
Y_t &= F(A_{Xt}X_t, A_{Lt}L_t) \\
&= \left[\alpha(A_{Xt}X_t)^{1-\rho} + (1-\alpha)(A_{Lt}L_t)^{1-\rho}\right]^{\frac{1}{1-\rho}},
\end{aligned}
$$

- Inputs and elasticity:
  - X_t: knowledge-intensive composite
  - L_t: unskilled labor
  - $1/\rho$: elasticity of substitution between inputs
- Firm maximizes $Y_t - P_t X_t - w_{Lt} L_t$ and earns zero profit

Knowledge-intensive good sector

- Competitive firm aggregates differentiated intermediate goods:

$$X_t = n_t^{-\frac{1}{\sigma-1}} \left(\int_0^{n_t} x_t(j)^{1-1/\sigma} \, \mathrm{d}j \right)^{\frac{1}{1-1/\sigma}}, \qquad \sigma > 1$$

- Components:
  - $x_t(j)$: input of variety j
  - n_t: measure of available varieties
  - $n_t^{-1/(\sigma-1)}$: separates market power from taste for variety
- Symmetry implies $x_t(j) = x_t$ and

$$X_t = n_t x_t$$

Intermediate good sector

- One unit of skilled labor produces one unit of intermediate good
- Firm j chooses price and quantity to maximize

$$d_t(j) = (p_t(j) - w_{Ht})x_t(j)$$

- Monopolistic competition implies

$$p_t(j) = \frac{\sigma}{\sigma - 1} w_{Ht}$$

- Firm distributes profit as per-share dividend:

$$D_t = d_t(j) = \frac{w_{Ht} x_t}{\sigma - 1}$$

R&D sector

- One unit of skilled labor devoted to R&D creates an_t new varieties
- Existing knowledge raises R&D productivity
- Each variety becomes new firm whose stock claims future monopoly profits
- Founder sells an_t shares at common ex-dividend price Q_t
- If innovation occurs, occupational indifference requires

$$\underbrace{an_t Q_t}_{\text{IPO proceeds}} = \underbrace{w_{Ht}}_{\text{production wage}}$$

- Thus Q_t prices ideas and protected knowledge

Saving and stock demand

- For type $i \in \{H, L\}$,

$$\begin{aligned} c^y_{it} + Q_t\theta_{it} &= w_{it}, \\ c^o_{i,t+1} &= (Q_{t+1} + D_{t+1})\theta_{it} \end{aligned}$$

- Unit EIS gives

$$c^y_{it} = (1-\beta)w_{it}, \qquad \theta_{it} = \frac{\beta w_{it}}{Q_t}$$

- Common SDF across types: wage cancels from intertemporal marginal rate of substitution

Preferences + derivation

Market clearing

Let $\phi_t \in (0, 1]$ denote skilled-agent share in production

- Skilled labor:

$$X_t = n_t x_t = \phi_t H$$

- Knowledge accumulation:

$$n_{t+1} = [1 + a(1 - \phi_t)H]\, n_t$$

- Stock market:

$$\underbrace{n_{t+1}}_{\text{supply}} = \underbrace{H\theta_{Ht} + L\theta_{Lt}}_{\text{demand}}$$

- Equilibrium reduces to determining ϕ_t

Equilibrium

Let $\zeta_t := A_{Xt}H/(A_{Lt}L)$ and $g(x) := (F_X/F_L)(x,1)$

Proposition

If innovation occurs, $\phi_t \in (0,1)$ uniquely solves

$$\frac{1}{aH} = \phi_t - 1 + \beta + \frac{\beta\sigma}{(\sigma-1)\zeta_t g(\zeta_t\phi_t)}.$$

Prices are

$$\begin{aligned} P_t &= p_t(j) = F_X A_{Xt}, \\ w_{Ht} &= \frac{\sigma-1}{\sigma} F_X A_{Xt}, \qquad\qquad w_{Lt} = F_L A_{Lt}, \\ Q_t &= \frac{w_{Ht}}{an_t}. \end{aligned}$$

No-innovation corner

Knowledge spillovers

Assumption

Factor-augmenting productivities satisfy

$$(A_{Xt}, A_{Lt}) = (A_X n_t^{\xi_{z_t}}, A_L n_t^{\lambda_{z_t}}),$$

with

$$\kappa := (\xi_u - \lambda_u)(\rho - 1) > 0, \qquad \lambda_u > \lambda_b = \xi_b.$$

- State u: uneven factor-augmenting spillovers and unbalanced growth
- State b: equal spillovers and balanced growth
- If $1/\rho < 1$, then $\kappa > 0$ is equivalent to $\xi_u > \lambda_u$

Innovation and knowledge growth

With CES production, equilibrium condition becomes

$$\frac{1}{aH} = \phi_t - 1 + \beta + \frac{\beta\sigma(1-\alpha)}{(\sigma-1)\alpha}\left(\frac{A_X H}{A_L L}\right)^{\rho-1} n_t^{(\xi_{z_t}-\lambda_{z_t})(\rho-1)}\phi_t^{\rho}$$

Proposition

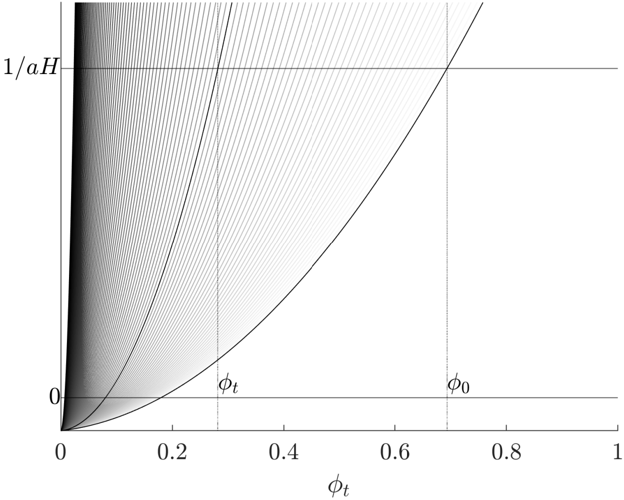
Conditional on continued state u, $\phi_t \downarrow 0$ and

$$\frac{n_{t+1}}{n_t} \to G_u := 1 + aH.$$

In state b, $\phi_t = \phi_b$ and

$$\frac{n_{t+1}}{n_t} = G_b := 1 + a(1-\phi_b)H < G_u.$$

Dynamics of ϕ_t



Inevitable stock price bubbles

Given active innovation,

$$\frac{D_t}{Q_t} = \frac{aH}{\sigma - 1}\phi_t$$

Theorem

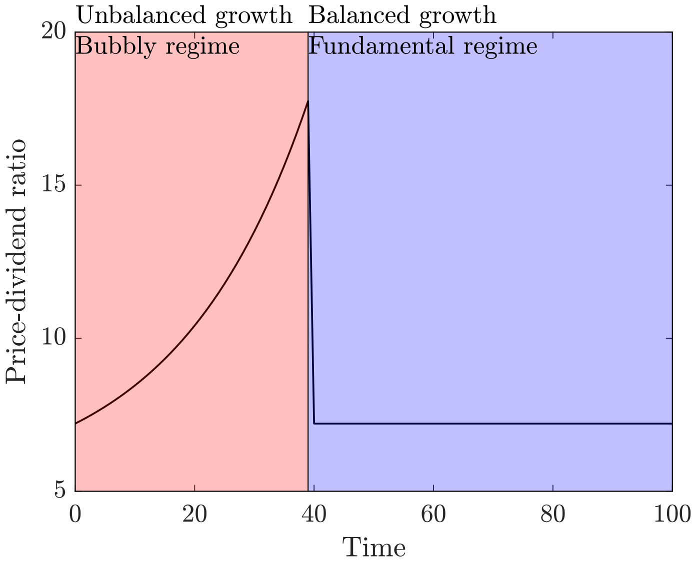
Suppose production is CES, innovation persists throughout state u, and $\gamma < 1$. Unique equilibrium:

1. *If $z_t = u$, $Q_t > V_t$ and Q_t/D_t grows asymptotically at exponential rate*
2. *If $z_t = b$, $Q_t = V_t$ and Q_t/D_t is constant*

Agents rationally anticipate transition to b and complete bubble collapse

▸ Proof ▸ Infinitely lived agents

Price-dividend ratio dynamics



Bubble-backed IPO proceeds

At fixed state-u history, write $Q_t = V_t + B_t$ and let innovators retain

$$Q_t^{\mathrm{IPO}}(\chi) = V_t + \chi B_t, \qquad \chi \in [0,1]$$

Proposition

As χ increases:

- $\phi_t(\chi)$ *decreases*
- *R&D employment and next-period knowledge increase*
- *skilled wage and skill premium increase*
- *current output and unskilled wage decrease*

One-period partial-equilibrium incidence experiment; all shares continue trading at Q_t

Details

Regime-transition consequences

Suppose economy switches from u to b at date t

Proposition

As transition occurs later,

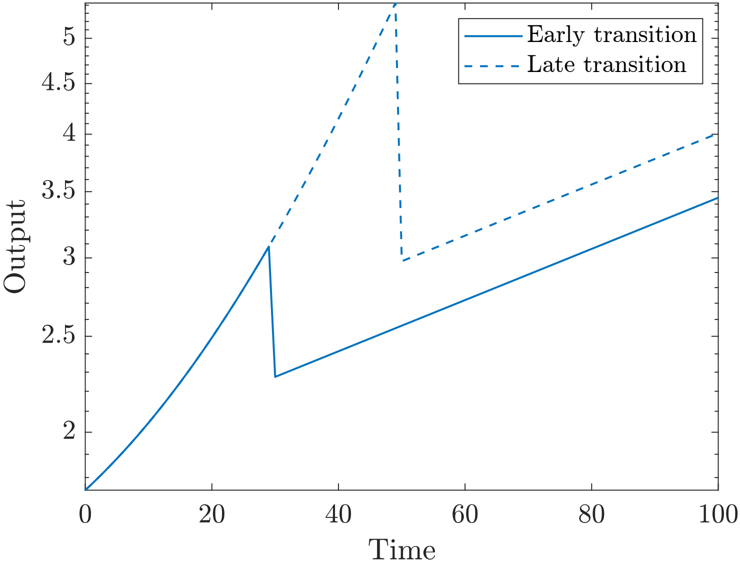
$$\frac{Y_t^b}{Y_{t-1}^u} \sim n_t^{\lambda_b - \lambda_u} \to 0, \qquad \frac{Q_t^b}{Q_{t-1}^u} \sim n_t^{\lambda_b - \lambda_u} \to 0.$$

If first transition occurs at τ, then for every fixed $k \geq 0$,

$$\begin{aligned} Y_{\tau+k} &= C_Y G_b^{\lambda_b k} n_\tau^{\lambda_b}, \\ w_{i,\tau+k} &= C_i G_b^{\lambda_b k} n_\tau^{\lambda_b}, \qquad i \in \{H, L\}. \end{aligned}$$

Later transition implies larger contraction but higher post-transition output and wage levels; post-transition skill premium unchanged

Output dynamics



Balanced growth: knife-edge

Proposition

Equilibrium price-dividend ratio is invariant to relative productivity A_{Xt}/A_{Lt} for every relative-productivity value if and only if F is Cobb–Douglas.
Within CES class, along growing path,

$$\frac{Q_t}{D_t} \text{ is constant} \quad \Longleftrightarrow \quad (\xi - \lambda)(\rho - 1) = 0.$$

Constant price-dividend ratio therefore requires either

- equal factor-augmenting spillovers, $\xi = \lambda$, or
- Cobb–Douglas production, $\rho = 1$

Balanced-growth restrictions eliminate relative-growth conditions generating bubble

Conclusion

- Temporary unbalanced growth can make collapsing bubble on dividend-paying stocks necessary
- Uneven GPT spillovers generate exponentially growing price-dividend ratio
- Bubble-backed IPO proceeds increase R&D and next-period knowledge in incidence experiment
- Later collapse more severe but leaves larger productive knowledge stock
- Balanced growth imposes knife-edge restrictions with strong asset-pricing consequences

Infinite horizon Bezos

Optimal consumption with log utility

- Let $R_{t+1} = (Q_{t+1} + D_{t+1})/Q_t$ be gross return
- Saving problem equivalent to

$$\max_{0<c<e_t} (1-\beta)\log c + \beta \log(e_t - c) + \text{constant}$$

- Hence

$$c_t^y = (1-\beta)e_t, \qquad Q_t = \beta e_t$$

Endowment economy
Stock demand

Stochastic discount factor

- One-period SDF:

$$m_{t\to t+1} = \frac{\beta}{1-\beta}\frac{c_t^y (c_{t+1}^o)^{-\gamma}}{\mathrm{E}_t[(c_{t+1}^o)^{1-\gamma}]}$$

- Using equilibrium consumption,

$$\begin{aligned} m_{t\to t+1} &= \frac{\beta}{1-\beta}\frac{(1-\beta)e_t(\beta e_{t+1}+D_{t+1})^{-\gamma}}{\mathrm{E}_t[(\beta e_{t+1}+D_{t+1})^{1-\gamma}]} \\ &= \frac{Q_t(Q_{t+1}+D_{t+1})^{-\gamma}}{\mathrm{E}_t[(Q_{t+1}+D_{t+1})^{1-\gamma}]} \end{aligned}$$

Return

Bubble-characterization proof

- Define

$$S_t := \pi^t \, \mathrm{E}_0[m_{0\to t} Q_t \mid \tau > t]$$

- Equilibrium SDF substitution gives

$$S_t = Q_0 \prod_{s=1}^{t} \frac{1}{1 + D_s^u / Q_s^u + \frac{1-\pi}{\pi} (\Pi_s^u / Q_s^u)(\Pi_s^b / \Pi_s^u)^{1-\gamma}}$$

- Writing denominator as $1 + a_s$, inequalities

$$1 + a \leq e^a, \qquad \prod_{s=1}^{t} (1 + a_s) \geq 1 + \sum_{s=1}^{t} a_s$$

- Hence $\lim_t S_t > 0$ if and only if $\sum_s a_s < \infty$, equivalent to theorem's two conditions

General EIS

- $\mathcal{R}_t(X)$: conditional CRRA certainty equivalent for positive X

$$\mathcal{R}_t(X) = \begin{cases} \mathrm{E}_t[X^{1-\gamma}]^{\frac{1}{1-\gamma}} & \text{if } \gamma \neq 1 \\ \exp(\mathrm{E}_t[\log X]) & \text{if } \gamma = 1 \end{cases}$$

- Let $\psi \neq 1$ denote EIS and define $x_t = Q_t/e_t$, $d_t = D_t/e_t$, and $G_{t+1} = e_{t+1}/e_t$; then

$$\frac{x_t^{\psi}}{1-x_t} = \left(\frac{\beta}{1-\beta}\right)^{\psi} \left[\mathcal{R}_t\left(G_{t+1}(x_{t+1}+d_{t+1})\right)\right]^{\psi-1}$$

- Under bounded growth and uniform interiority:
  - uniformly interior equilibrium exists
  - same primitive relative-growth conditions characterize bubbles
  - complete collapse still requires $\gamma < 1$
- For $\psi > 1$, greatest equilibrium is unique uniformly interior equilibrium; other equilibria may approach zero-price boundary

Economic interpretation

No-innovation corner

- Innovation occurs if

$$\frac{1}{aH} < \beta + \frac{\beta\sigma}{(\sigma-1)\zeta_t g(\zeta_t)}$$

- If condition fails, $\phi_t = 1$ and

$$Q_t = \frac{\beta(Hw_{Ht} + Lw_{Lt})}{n_t}$$

- Either way, ϕ_t depends on time only through relative effective productivity

$$\zeta_t = \frac{A_{Xt}H}{A_{Lt}L}$$

Return

Full-model bubble-necessity proof

- When innovation occurs,

$$\frac{D_t}{Q_t} = \frac{aH}{\sigma - 1}\phi_t$$

- In state u, ϕ_t decreases at exponential rate, so

$$\sum_t \frac{D_t^u}{Q_t^u} < \infty$$

- At transition,

$$\frac{Q_t^b}{Q_t^u} \sim n_t^{\lambda_b - \lambda_u} \to 0$$

- Together with $\gamma < 1$, this verifies

$$\sum_t \left(\frac{\Pi_t^b}{\Pi_t^u}\right)^{1-\gamma} < \infty$$

- Endowment-economy theorem then implies $Q_t > V_t$ in state u

Bubble capture and occupational choice

- Holding n_t, (A_{Xt}, A_{Lt}), V_t, and B_t at benchmark values, occupational choice satisfies

$$a n_t Q_t^{\mathrm{IPO}}(\chi) \leq w_{Ht}(\phi_t(\chi)),$$
$$(1-\phi_t(\chi))\left[w_{Ht}(\phi_t(\chi)) - a n_t Q_t^{\mathrm{IPO}}(\chi)\right] = 0$$

- Unique $\phi_t(\chi)$ decreases with χ, so

$$\widetilde{n}_{t+1}(\chi) = [1 + aH(1-\phi_t(\chi))]\, n_t$$

increases with fraction of bubble component captured by innovator

- Comparison excludes financial-market clearing, wealth effects, and fundamental-value recomputation

Return

Infinitely lived agents

Use demographic structure from Weil (1989):

- every cohort lives forever
- new skilled and unskilled cohorts enter at each date
- agents supply labor only at birth, then consume from financial wealth
- production, innovation, and regime process unchanged
- Recursive utility is

$$U_{i,\tau,t} = c_{i,\tau,t}^{1-\beta}\, \mathrm{E}_t[U_{i,\tau,t+1}^{1-\gamma}]^{\frac{\beta}{1-\gamma}}$$

- Unit EIS implies each agent consumes fraction $1-\beta$ of wealth

Return Result

Bubble necessity with infinitely lived agents

Theorem

Under active innovation and admissible continuation values, aggregate allocation, stock price, continuation-value solution, and SDF are unique. Furthermore:

1. *In state u, $Q_t > V_t$ and Q_t/D_t grows asymptotically at exponential rate*
2. *In state b, $Q_t = V_t$ and Q_t/D_t is constant*

- Each agent satisfies individual transversality condition
- Existing cohorts run down stock positions; new cohorts replenish aggregate stock demand with labor income
- Finite lives unnecessary; continual cohort entry remains essential

Return

Industrial bubbles

"The ones that are industrial are not nearly as bad. They could even be good because when the dust settles and you see who are the winners, society benefits from those inventions."

Jeff Bezos, Italian Tech Week, October 3, 2025
Video recording

Main results
Conclusion

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